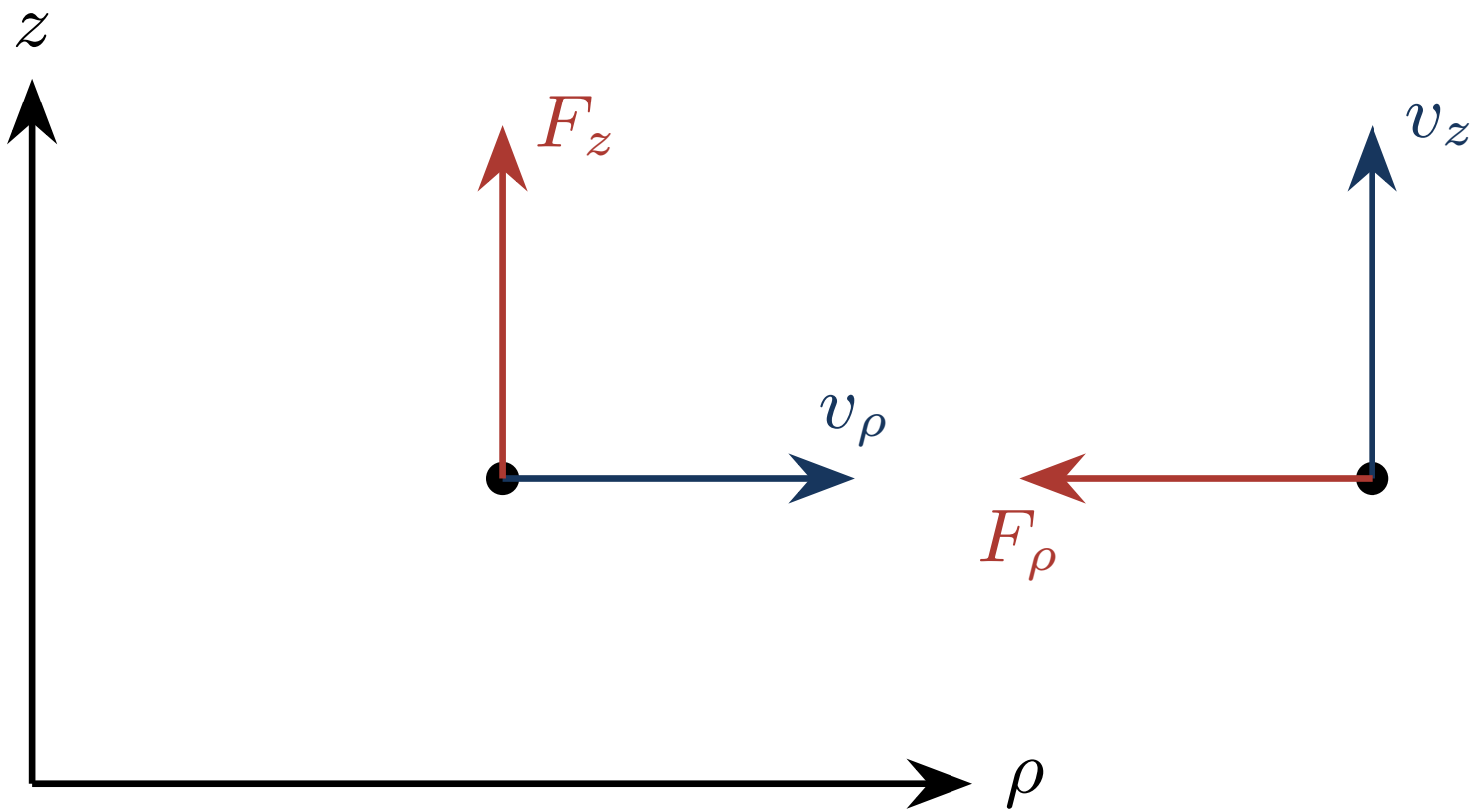
$$qI > 0, \;\; F_\theta = 0$$



$$F_z = q\kappa v_\rho/\rho, \;\; F_\rho = -q\kappa v_z/\rho$$